



THE LONDON SCHOOL
OF ECONOMICS AND
POLITICAL SCIENCE ■

September 2025 Exam

EC400

Dynamic Programming and Differential Equations

Suitable for all candidates

Instructions to candidates

This exam has nine questions. Answer all questions.

Please write your answers in dark ink (black or blue) only.

Only answers written in an answer booklet will be sent for marking.

Time Allowed	Reading Time:	<i>None</i>
	Writing Time:	<i>90 minutes</i>

You are supplied with:	<i>No additional materials</i>
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You may also use:	<i>No additional materials</i>
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Calculators:	<i>Calculators are not allowed in this exam</i>
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EC400: Exam

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To explain consumption-saving decisions of households and volatile asset prices, economists sometimes appeal to habits. In particular, one commonly used approach is to assume that household utility depends on its consumption C_t relative to some benchmark H_t :

$$\sum_{t=0}^{\infty} \beta^t \log(C_t - \theta H_t), \quad \text{where } 0 < \theta < 1.$$

For concreteness, assume that the last-period consumption is used as a benchmark $H_t = C_{t-1}$. The household then solves a pie-eating problem maximizing utility subject to the budget constraint

$$B_{t+1} = B_t - C_t,$$

where $B_t \geq 0$ is the size of the remaining pie. The initial values B_0 and H_0 are given.

1. (10 marks) Substitute $H_t = C_{t-1}$ into the objective function, write down the Lagrangian and take the first-order conditions.
2. (10 marks) Derive the intertemporal optimality condition (Euler equation). Using analogy with a finite-horizon model, write down the transversality condition.
3. (12 marks) Go back to the formulation of the problem in terms of $\{B_t, H_t, C_t\}$. Which variables are states and which ones are controls? Find the steady state in this model. If the Blanchard-Kahn principle were satisfied, would the steady state be a source, a sink, or a saddle point (no calculations required)?
4. (10 marks) Write down the Bellman equation for the problem.
5. (12 marks) Conjecture that $V = \alpha + \gamma \log(B - \phi H)$. Solve for γ, ϕ (no need to solve for α) and find the policy function.

6. (10 marks) Rewrite the laws of motion for B_t and H_t for a period of arbitrary length and show that the continuous-time versions of the equations are given by $\dot{B}_t = -C_t$ and $\dot{H}_t = C_t - H_t$.
7. (12 marks) Define the discount factor ρ such that $\beta = e^{-\rho}$. Write down the Hamiltonian and the optimality conditions according to the maximum principle.
8. (12 marks) Guess and verify that co-state variables are constant in time and derive the (continuous-time) Euler equation in terms of C_t, H_t and their time derivatives.
9. (12 marks) Start with the discrete-time Bellman equation and derive the HJB equation for this problem.